

KIRAN SHAY

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Computer Science & Neuroscience researcher applying machine learning to biological signal analysis — from wearable tremor classification to RL-based reasoning models — seeking full-time neurotech roles.

EDUCATION

Johns Hopkins University

Bachelor of Science in Computer Science and Neuroscience | Minors: Applied Mathematics & Statistics, Business

Aug 2023 – May 2027 (expected)

- Cumulative GPA: 3.90
- Relevant Coursework: Data Structures, Algorithms, Machine Learning, Probability & Statistics, Linear Algebra

RESEARCH EXPERIENCE

Antidepressant Use in Pregnancy — Drug Side Effect Research

Research Software Engineer Intern, JHU Data Science and AI Institute (DSAI) (PI: Dr. Casey Taylor) | Summer 2025 – Present

- Studied adverse effects of antidepressant use during pregnancy using the All of Us (AoU) research database, with the REACH pregnancy cohort used as a verification/replication dataset
- Continuing the analysis and manuscript this semester; preparing the paper for conference submission

Delineo: Infectious Disease Simulation Research

Simulation Team Engineer | JHU Dept. of Computer Science | May 2025 – Dec 2025

- Contributed to the core simulation engine modeling stochastic disease spread across communities
- Improved computational efficiency by profiling code and optimizing NumPy-based pipelines
- Refined geographic modeling by smoothing contiguous zones to improve model continuity and accuracy

Probabilistic Lineup Optimization for Baseball

Research Lead | Quantitative Modeling, Simulation, Conditional Probability | Dec 2024 – Dec 2025

- Modeled optimal batting order strategies using conditional probability matrices, Markov-style transitions, and stochastic simulation; developed BRP, a custom expected-run-contribution metric
- Developed and deployed full-stack infrastructure to support the simulation platform
- Received the 2025 Johns Hopkins Whiting School of Engineering Design Day Digital Vanguard Award

TremorML: Time-Series Classification of Neurological Disorders

Research Lead | Machine Learning, Signal Processing, Time-Series Modeling | Jan 2025 – Mar 2026

- Built a machine learning pipeline to differentiate tremor types (e.g., Parkinsonian vs. essential tremor) from wearable time-series sensor data
- Implemented and evaluated XGBoost, random forests, and deep neural networks, improving minority-class performance with SMOTE class balancing

LEADERSHIP & EXPERIENCE

Affordable Tutoring Solutions Inc.

Co-Founder & Chief Technology Officer | Educational Accessibility | June 2023 – Present

- Designed and deployed full-stack web infrastructure for scheduling, curriculum delivery, and performance tracking
- Led development of personalized, data-informed academic curricula aligned with SAT/ACT standards

Kumon North America

Motivational Speaker / Student Representative | 2015 – Present

- Selected to speak at major Kumon student and instructor conferences across the U.S., Canada, and Mexico; representing Kumon North America at the 2025 Global Virtual Student Conference

TECHNICAL SKILLS & INTERESTS

Languages & Tools: Python, NumPy, XGBoost, PyTorch/TensorFlow

Methods: Machine learning, stochastic modeling, Monte Carlo simulation, Markov chains, time-series analysis, optimization

Additional Research: Implemented and compared PPO, DPO, and GRPO for verifiable-reward LLM reasoning tasks (HumanEval, GSM8K) — Spring 2025

Interests: Classical piano, travel, bodybuilding, logic puzzles, chess, sports analytics